7	(a)	The decay of radioactive nuclei is said to be *random* and *spontaneous*. Explain what is meant by the radioactive decay is *random* and *spontaneous*.	
		random: …………………………………………………………………………………………	
		………………………………………………………………………………………………	
		………………………………………………………………………………………………	
		spontaneous: ……………………………………………………………………………………	
		………………………………………………………………………………………………	
		………………………………………………………………………………………………	[2]
	L1	**Random:** It is **impossible to predict which particular <u>radioactive</u> <u>nucleus</u> in a given sample will decay next and when it will decay,** **even though <u>any nucleus</u> has a constant probability of decay per unit time.** ** Reference to 'nucleus' is required.* ** Full definition, including mention that any nucleus has a constant probability of decay per unit time, is necessary for full credit.* **Spontaneous:** The decay **occurs without external stimuli and the rate of decay is unaffected by environmental factors such as temperature and pressure.**	B1 B1
	(b)	A Geiger-Müller counter was used to measure the count rate *C* of a radioactive source over several years. The readings were recorded and used to obtain the graph in Fig. 7.1.	

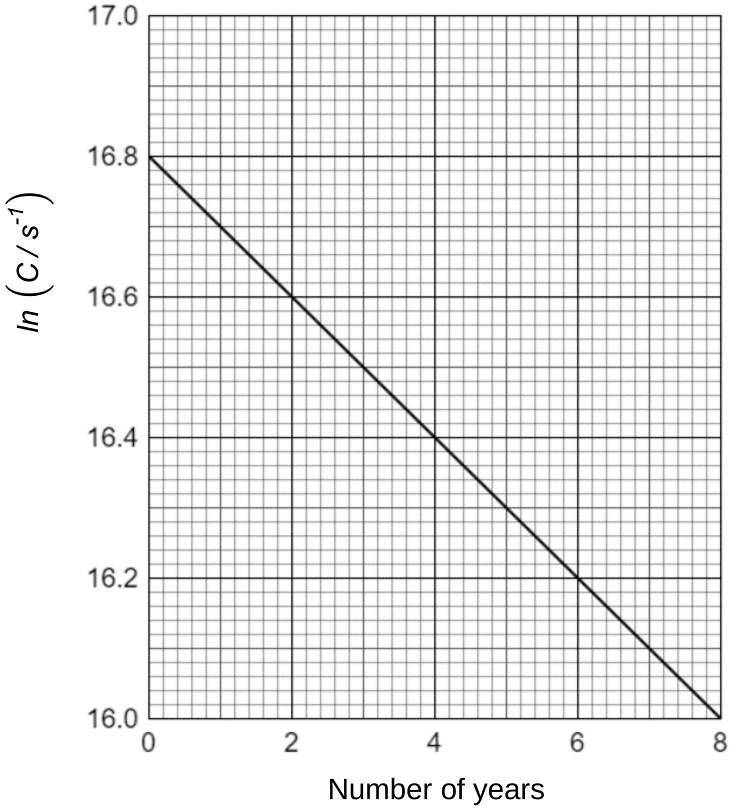

Fig. 7.1

(i) Using Fig. 7.1, determine the decay constant.

decay constant = ……………..………………… s^{-1} [3]

L2

$C = C_0 e^{-\lambda t}$

Linearise:

$\ln C = \ln C_0 - \lambda t$

$\ln C = (-\lambda)t + \ln C_0$

The gradient of Fig. 7.1 is $-\lambda$. **B1**

$-\boldsymbol{\lambda} = \dfrac{\mathbf{16.8 - 16.2}}{\mathbf{0 - 6}} = -\,\mathbf{0.1}\ \boldsymbol{year^{-1}}$ **M1**

$\lambda = 0.1\ year^{-1} = \dfrac{0.1}{365 \times 24 \times 3600} = 3.1710 \times 10^{-9} = \mathbf{3.17 \times 10^{-9}}\ \boldsymbol{s^{-1}}$ **A1**

(ii) Determine the half-life of the radioactive isotope.

half-life = ……………………………….… s [1]

L1

$t_{1/2} = \dfrac{\ln 2}{\lambda} = \dfrac{\ln 2}{3.1710 \times 10^{-9}} = \mathbf{2.19 \times 10^{8}}\ \boldsymbol{s}$ **A1**

(c) Describe what an experimenter would do in the measurement of the half-life of the sample to reduce the effect of

(i) the random nature of the radioactivity decay process,

			…………………………………………………………………………………………………	
			…………………………………………………………………………………………………	
			…………………………………………………………………………………………………	[1]
		L3	**Measure the count rate for a longer period of time**, in the process, **averaging out the random fluctuations**. OR Use a **sample with greater number of radioactive nuclei to increase the measured count rate,** thereby reducing the percentage error in the counting.	**B1**
		(ii)	the background radiation.	
			…………………………………………………………………………………………………	
			…………………………………………………………………………………………………	
			…………………………………………………………………………………………………	[1]
		L2	First **measure the background radiation count rate <u>in the absence of</u> the sample**. Then **subtract the background radiation count rate from the measured count rate <u>in the presence of</u> the sample**.	**B1**

[Total: 8]

Section B

Answer **one** question from this Section in the spaces provided.

8	A mass spectrometer separates charge particles based on mass-to-charge ratio so that the